

Enterprise modelling the telecommunications industry

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The telecommunications industry is becoming both more co-operative and more competitive and this has dramatic consequences on the character of operational support. Operational activities make up the majority of telecommunications companies, and investment in these activities is huge. Dynamic markets and emerging new roles and supply chains stress the design of such operational systems and processes, some of which can trace their origins back to the monopolistic days of the past. This paper presents preparatory work in the search for new architectural tools to prepare operational support for the next millennium. The paper provides a definition for a value-adding enterprise and shows how conversation theory can help analyse the interaction of roles within an enterprise model. Such models can describe the interaction between responsibilities within enterprises and, in future, the required operational support systems, activities, data and processes.

1. Introduction

The telecommunications market, like other markets, is becoming a world of competitive co-operation. An understanding of how the roles and responsibilities fit within the overall supply chain is essential for the definition of the processes, the systems and their interfaces. Operational support is vital for a service company, and the investment is large. The vast majority of a telecommunications company's resource is involved in the support of the services that it offers to the market.

Within the traditional monopolistic model of telecommunications all aspects of the service were supplied and supported by the same enterprise or business. There was no need to consider separate systems and processes for different parts of the overall solution. Many factors have eroded this situation. Regulatory harmonisation measures, the emergence of new technologies and networking paradigms, and shifts in the expectation of the market mean that today a number of segmented roles exist. Operational support will mean different things to each different role. Different supply chains, built from market players taking on these roles, imply different support requirements and interactions. The traditional functionality is now being split along business lines and co-operation is now needed to tie this functionality together to provide the end customer with the service they want.

If our analysis is correct, it suggests a set of roles that could be adopted in order to provide a better understanding of the value chain, one based on a model of the value-

adding enterprise presented in this paper. It requires three top-level responsibilities — those related to scoping, resourcing and delivery. The supply chain consists of concatenated value-adders with interconnected resourcing and delivery responsibilities. To understand the supply chain, the responsibilities involved in each value-adding role need to be understood — who is supplying what to whom.

2. What is enterprise modelling

There are many different flavours of enterprise modelling. Reviews of enterprise modelling efforts [1—4] show that much of it is aimed at understanding and improving business processes, although other specific purposes for producing enterprise models, such as activity-based costing, ISO9000 and systems analysis, are quoted.

This paper illustrates enterprise modelling as a means of understanding the problem space rather than populating the solution space. Rather than aiming to integrate different IT solutions, it is used to decompose large, complex systems that have evolved within a 'single enterprise culture'.

Enterprise modelling is an analysis technique that can be used to model systems, businesses or organisations. The technique is flexible as it focuses on the roles and responsibilities of enterprises and not just how these roles and responsibilities are undertaken. Unlike other analysis

techniques, it concentrates on exploring the problem, not describing the problem with a view to its solution. Understanding the roles within an enterprise helps shape whatever resources are used to fulfil them, be they people, processes or hardware and software systems.

Responsibilities that have strong links between them are grouped to form roles. Enterprise modelling can then be used to explore how these roles can be split between different enterprises. The use of the value-adding concept gives criteria for the formation for such enterprises. Alternative models can be analysed by adding responsibilities or by moving a responsibility from one enterprise to another.

Communication between the resultant enterprises is then examined. This analysis between the resultant enterprises is called conversational analysis. The use of conversational analysis helps investigate how these enterprises co-operate to meet a market need.

2.1 What is an enterprise

An enterprise has a number of responsibilities and owns the resources required to fulfil them. The scope of the analysis will determine the rationale behind identifying enterprises within the model. A particular rationale is discussed in section 3. An enterprise model is abstract, it describes responsibilities not who or what fulfils them. In reality the roles and responsibilities are performed by players, e.g. companies. Enterprises can map to players or departments within a larger player. However, there need be no direct mapping, and responsibilities within an enterprise may be spread across many departments within a player, or across a number of players. For example, the responsibility for revenue collection may form part of the responsibilities of an invoice management department within one company, but within another company the same responsibility may be split between an invoicing department and a third-party debt-collection agency.

2.2 Conversational analysis

Within enterprise modelling, any items that are owned by enterprises are called resources. Ownership is meant in a broad sense; people employed by the enterprise would be considered resources. The interactions between enterprises, called conversations, are modelled by the exchange of instruments. Instruments do not dictate the technology used to convey them. An instrument called receipt could be paper, electronic or a plastic token. At this level, the enterprise model does not concern itself with such detail.

Conversations can be assessed in terms of their number, the complexity and number of instruments, etc, but are also assessed using the following four criteria.

- Mutuality

Mutuality is the responsibility one party undertakes on behalf of the other. A car dealership has a significant responsibility in ensuring that the car sold to the customer meets their requirements. To this end it may offer test-drives, warranties, etc. Therefore, the retailer to customer conversation has high mutuality. The same car sold at auction is sold as seen, and buyers have no recourse if they are unsatisfied with the car or if the car immediately fails. The auction house to customer conversation has low mutuality. The mutuality on the buyer in both cases is low as all they need to do to satisfy the other party is to pay, and the seller in both cases will not let them have the car until they do. Therefore, the above two examples could be described as having high-low and low-low mutuality respectively.

- Significance

Significance is the importance of the conversation to both parties. Returning to the car-buying scenario, the significance to the buyer of the buying conversation will usually be high. Buying a car is one of the biggest investments people make. To the auctioneers that may sell thousands of cars per week, the conversation will have lower significance.

- Control

This is the degree of control each party has within the conversation, i.e. which parties can initiate, suspend, leave and rejoin the conversation as it passes through its phases. When buying a car from a dealership, customers hold most of the control. They approach the dealership, make an offer when/if they deem it appropriate and can pull out at any stage. In the auction, the situation is different. The cars are offered for sale for a very limited period (i.e. a few minutes). Potential buyers must bid a price within guidelines set by the auctioneers. Buyers cannot suspend the negotiation to go away and think about it. When the hammer falls, the contract is made; bids cannot be withdrawn.

- Information

This relates to the information both parties require in order to meet their responsibilities regarding the conversation and how this is distributed between the two parties. The car dealer holds the information about the products for sale, the buyer holds the requirements e.g. price, performance. Certain aspects of this information need to be communicated from one party to the other during the sales conversation.

2.3 From enterprises to players

A model of an enterprise is an abstract concept, in that it represents a grouping of responsibilities that are coupled, regardless of how those responsibilities are performed. BT is not an abstract concept; it is a significant player within the telecommunications market. Not only this, but it adopts many roles within that market. So what is the relevance of these models to BT (or any other player)?

BT's size and ability to provide vertically integrated services are strengths, but also can constitute a weakness. There can be a tendency to ignore the different aspects of a service and develop large and tightly integrated, 'stove-piped' systems. This has a number of drawbacks:

- lack of clarity as to what is being offered to the customer — whether it is network capability or a service, and which of the elements adds the most value,
- lack of reuse — if a service is bundled with the associated network and access capabilities, it may be difficult to unbundle these elements to re-bundle them within other services (this is especially true if the operational support elements have been similarly bundled),
- lack of flexibility — a service provider may wish to exploit particular network capability, but cannot do so without using the associated service, e.g. it is impossible to use the telephony network without using the telephony service.

A model which allocates service components to separate enterprises, even if only hypothetically, helps to identify what is being offered to the customer and what value each enterprise is adding.

2.4 From players to systems

Where there are commercial interfaces between systems, the nature of the interface and the responsibilities undertaken on either side of that interface are clear. They may even be explicitly stated within a contract. When system interfaces are internal, there can be a lack of clarity about the role a system plays within a supply chain.

As systems evolve, their initial role can be corrupted as designers make pragmatic, short-term decisions as to which element of a system should be modified to fulfil a requirement. When there is a commercial interface, such dilution of system definition is less likely to occur as separate commercial organisations are not going to take on additional responsibilities unless there is a business case to do so.

An enterprise model illustrates how different elements of large systems are related in a value chain. It can be used

to clarify roles and the interfaces between them to produce the same lack of ambiguity that would exist if the systems existed within separate commercial organisations.

2.5 Enterprise modelling diagrams

Figure 1 is an enterprise model showing the car-buying conversation. Other conversations also take place between the enterprises (e.g. pre-sales, post-sales support), but only the sales conversation is shown as an example.

As with most modelling techniques, the same scenario can be represented in a number of different ways and at one or more of a number of different levels of granularity. There is no right way, but some models are more useful than others. The model shown in Fig 1 is far from complete and is merely illustrative.

The instruments that are internal to the enterprises are not shown, but these would include instruments such as a spending limit to be passed between the 'financer' and the 'requirements specifier' and an authorisation to release the car between 'revenue collection' and 'sales'. The model illustrates the following points.

The customer, although frequently one person, includes a number of roles. A failure of any of these roles could lead to a failure in the overall conversation — for example, if the requirements are not correct, this will lead to the purchase of an inappropriate vehicle. There are examples where the 'requirements specifier' and the 'car user' are distinctly different roles, e.g. the purchase of a fleet of cars by a fleet manager. In this instance, the requirements will be more influenced by such factors as running costs and depreciation and less by the user's personal tastes.

In a large sales enterprise, the sales agent may include a large number of people; for a sole trader, a single person fulfils both sales and revenue collection. This reinforces the point that responsibilities (or groups of responsibilities) are not meant to represent job descriptions.

3. The value-adding enterprise

When forming enterprises from groups of roles, there must be some justification for their formation. A data-centric approach may be adopted, forming enterprises from responsibilities that share the same data resource. However, a much more interesting and useful approach is a value-centric approach. Systems, people and resources all cost money; to justify this expense they must add value in some way. Therefore value-adding enterprises can be formed, analogous to separate businesses. The model shown in Fig 2 embodies the premise that to add value, an enterprise is constituted with three fundamental roles — scoping, resourcing and delivering; these will be described shortly.

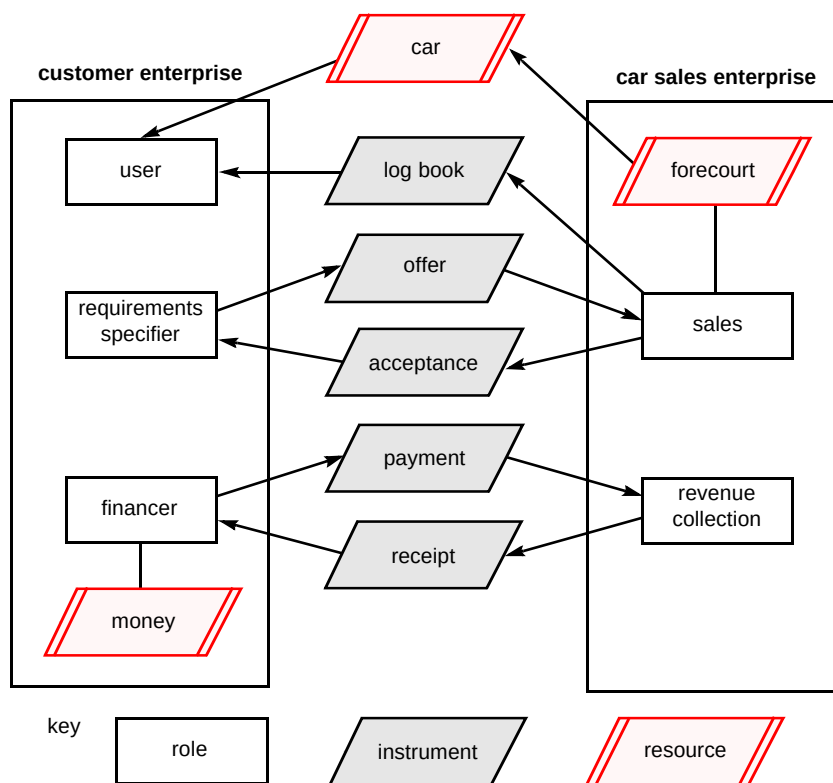


Fig 1 Car sales enterprise diagram.

These enterprises may align to businesses (players) in the real world, but it is more likely that a player will consist of a number of enterprises. The size of a particular enterprise will depend on the level at which the model is operating.

All value-adding enterprises add value to something and therefore follow the same basic model as shown in Fig 2.

Enterprises can add value in a number of ways:

- by changing the nature of the assets purchased, e.g. manufacturing,
- by providing economies of scale so allowing many users to share the same resource, e.g. telecommunications services,
- by collocating products supplied by many manufacturers, e.g. retailing,
- by selling knowledge and expertise, e.g. brokers and consultants.

What all these value-adders have in common is the need to procure resources from suppliers and deliver to the customer. It can be seen that the enterprise operates within two markets, as a customer in the market for the resources it procures and as a supplier in the market for the goods and services it provides. The enterprise needs to understand both these markets.

The model (Fig 2) identifies three areas of responsibility (scoping, resourcing, and delivering) that exist within every value-adding enterprise. Each of these areas has the responsibility to interpret a range of information in order to generate an internal instrument for the other two areas to interpret.

• Scoping

Determining the strategy of the enterprise — what it is going to supply and to whom. Scoping a car dealership, for example, is to decide what quality, quantity and make(s) of car the car dealership will sell.

• Resourcing

Procuring and managing the required resources in order to deliver the business, both in terms of resources that are passed on to the customer (i.e. stock) and resources that are required to run the enterprise (e.g. premises).

• Delivering

Delivering the business, including pre-sales, sales and post-sales support and revenue collection.

In order to add value, the value provided by 'delivering' must be greater than that consumed by 'resourcing'. These roles are described in more detail below.

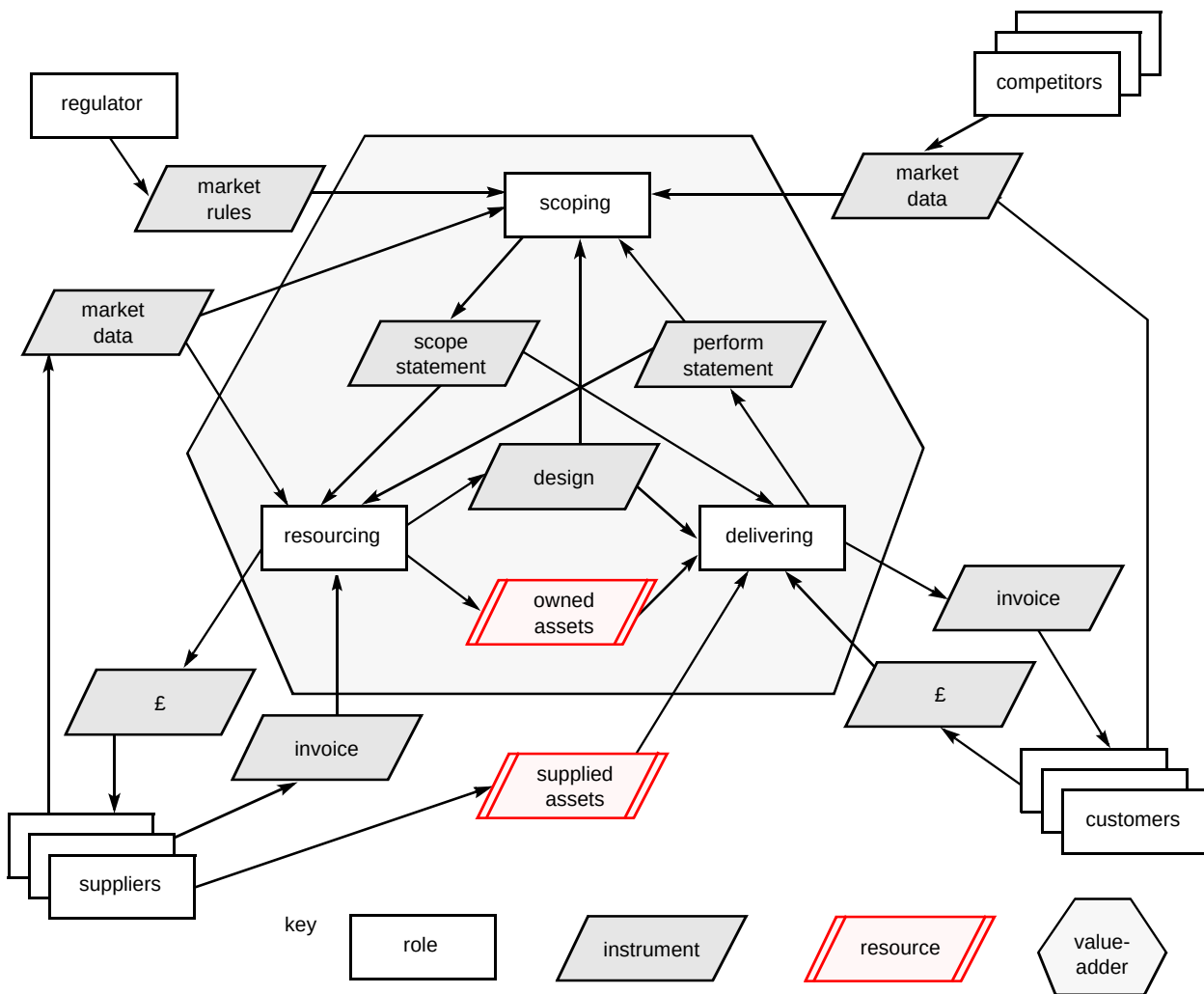


Fig 2 The value-adder model.

Table 1 shows one example of how departments within organisations might be mapped to the roles.

Table 1 Mapping of departments to responsibilities.

Scoping	Resourcing	Delivering
Board of Directors Strategy and planning Marketing Research	Purchasing Supply management Recruitment Construction of infrastructure	Sales Invoice management Support Production

Scoping, see Fig 3, corresponds to the responsibility to define and review the set of specific offers that the enterprise will make to the market, i.e. it defines the nature of the business. As with the other internal responsibilities, this involves the interpretation of both external and internal information. External sources (market data, comprising market conditions and market offering in Fig 3) relate to the demand and the offers available on the market. If market offers are considered relevant to the operation of the value adder, then they are either interpreted as competition, which must be responded to, or as opportunities which may be

exploited in the delivery or marketing of current or possible future offerings.

The internal information interpreted by the scoping responsibility is generated through the exercise of the other two internal responsibilities, resourcing and delivering. The 'perform(ance) statement' will detail how the goods and services offered are performing, not just functionally, but also in terms of sales, revenue, etc. This information will be used to manage the portfolio of goods offered by the enterprise.

The 'design statement' details how the goods and services that are offered have been implemented; scoping will use this information to determine the feasibility of using current infrastructure to satisfy future market needs.

The internal instrument 'scope statement' documents the set of products or services that the enterprise intends to deliver, the market in which it expects to operate and its expectations regarding the conditions it will encounter and performance it aims to achieve.

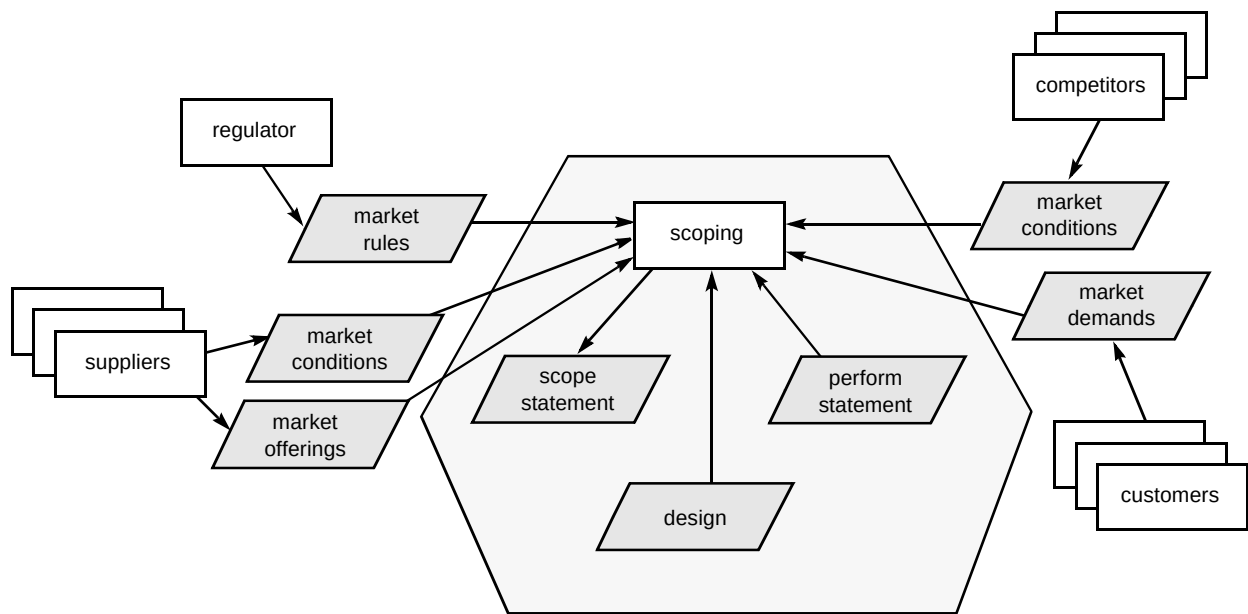


Fig 3 Scoping.

Resourcing, see Fig 4, corresponds to the responsibility to define, procure and deploy the resources required to achieve the delivery of the business. The internal information resources that are interpreted in discharging this responsibility comprise the scope statement and the record of performance (perform statement) of the existing organisational structure and infrastructure. The external information resources correspond to the market offers of suppliers (market offerings) at supporting levels in the value chain. Three classes of procurement may be distinguished.

- **Raw material and component procurement** is the procurement of items that will be incorporated in the output of the value-adding enterprise and sold on, e.g. integrated circuits procured for the manufacture of mobile telephones. The ownership of these items is eventually passed on to the customer.
- **Service resource procurement** is the procurement of the resources that are required to create a service offering; the ownership of these resources does not pass to the customer, e.g. telecommunications switches.
- **Business systems procurement** is the procurement of assets that support the business rather than form part of the product or service offered to the customer, e.g. office automation systems, accountancy systems.

Exploited resources (either 'owned assets' or 'supplied assets') are positioned on the boundary of the value-adding enterprise indicating that some remain outside the ownership of the procurer, some are temporarily owned and some are owned, depreciated and written off within that enterprise. All of these resources, expressed in terms of rights of ownership and rights to use, fall into the category

of assets and are evaluated on the basis of the return on investment that they yield to the value-adding enterprise.

Resourcing also includes the right to generate tenders, orders and registrations, and the obligation to pay for supplied resources that figure in the external conversations with suppliers.

Having procured the required assets, resourcing has the responsibility to produce the design statement that describes what has been constructed/manufactured. It includes primarily the information that the delivering responsibility requires to configure and sell/operate the offered goods or services.

Delivering, see Fig 5, interprets the scope statement and the design, and uses, operates or consumes the resources ('owned assets' and 'supplied assets') that correspond to infrastructure and inputs in order to deliver the value-adding business. A key aspect of this is the responsibility for dealing with the customers, including pre- and post-sales support as well as sales and the associated revenue collection. Delivery has the responsibility for documenting the delivery performance. The production of the perform statement completes the feedback loop. It enables scoping to determine the extent to which the aims within the scoping statement have been met and informs resourcing as to the performance of their chosen suppliers.

4. The telecommunications service provision enterprises

A consideration of the resources consumed that enable the supply of telecommunications services, and the

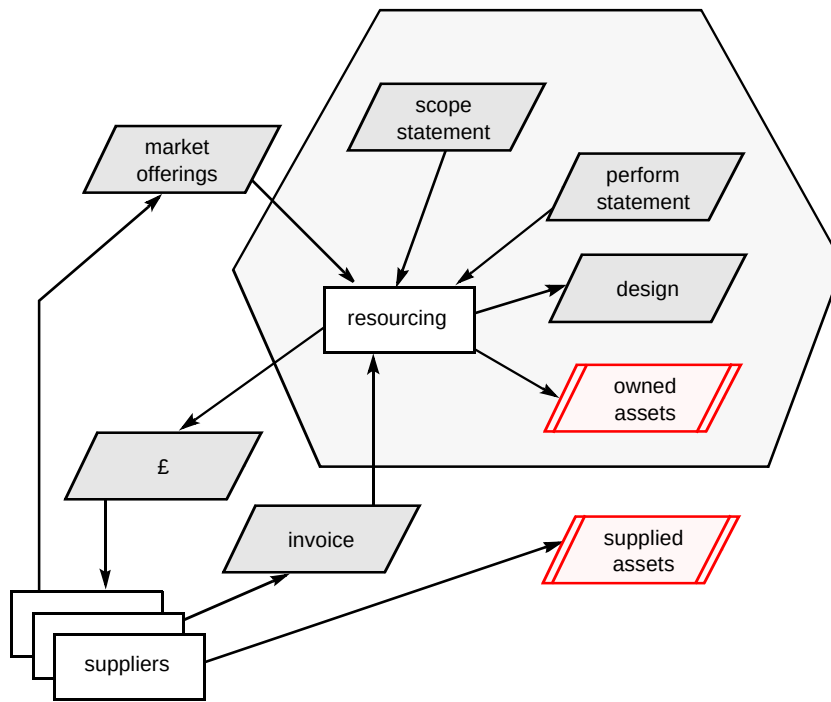


Fig 4 Resourcing.

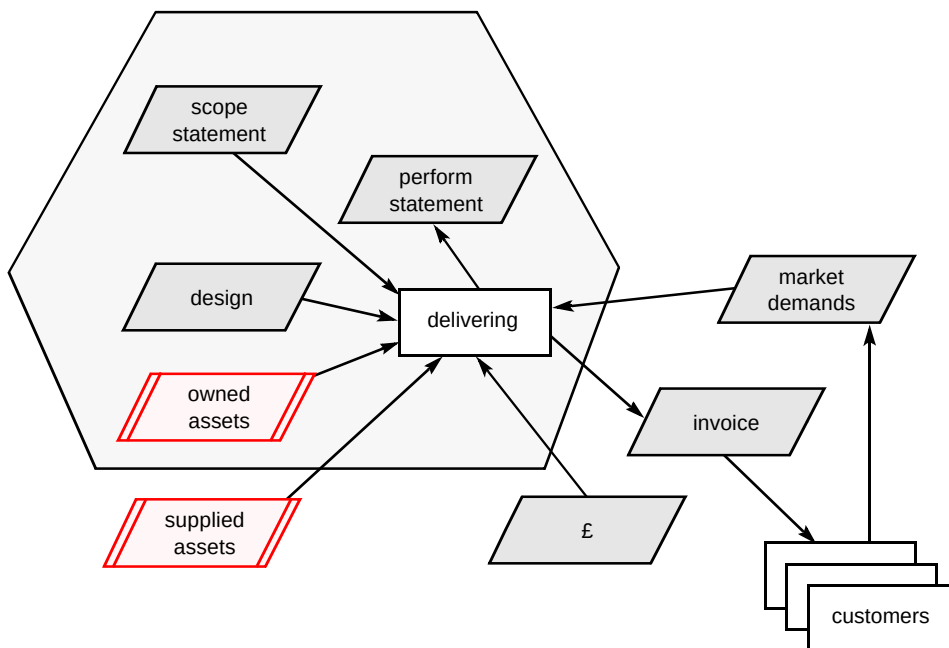


Fig 5 Delivering.

value added to these resources, using our enterprise modelling methodology has identified three distinct sets of responsibilities (plus the customer) within the telecommunications service supply chain. These are the 'access connector', the 'network connector' and the 'service provider'. The main roles are described below.

- Customer

To identify, procure and pay for services used to meet their requirements — the customer has the three roles

that were detailed in Fig 1 (user, specifier and financier).

- Access connector

To provide the physical connectivity between the customer premises and the network point of presence.

- Network provider

To carry data between addressed network connection points.

- Service provider

To provide a service (utilising the underlying network) that meets the customers' requirements. Clearly not all service providers use a telecommunications network; this constraint only applies to service providers within this model.

The definition of these responsibilities is not meant to imply that there will be three different companies within the supply chain, but that there are three distinct roles that can be adopted within the supply chain.

4.1 The access value-adder

The access connector provides a point-to-point link from the customer to the network. In effect, the access simulates the presence of the network to the customer (and vice versa). Within telephony, a copper pair from the home to the local exchange generally provides access. For mobile telephony, access is provided by the radio path from the handset to the mast and then the point-to-point link from the mast to the switch. Access connectors add value by enabling many users to share the use of common resources to connect them to a network point of presence. The resources involved include the access infrastructure (e.g. masts, ducts), and the plant, equipment and people required to establish and maintain the access links.

4.2 The network value-adder

The network connector, within the enterprise model, provides bit transport services. This excludes services such as e-mail, telephony and number translations which are provided by service providers. This is difficult to rationalise with the current PSTN, as the network only supports one service, i.e. telephony¹. However, this model is focused on understanding the future supply chain, where IP-based networks allow many services to be operated on the same network.

As with the access connector it is the sharing of a common resource that generates the value. For the network connector the resource is the telecommunications switches and the transport links between them. With public networks, these resources are used by many millions of people, and therefore the huge investment in network infrastructure can be recouped at a relatively low cost per user.

4.3 The service provider value-adder

The service provider role covers many possible services and indeed may involve the interaction between a number of service providers.

¹ In this regard telephony is regarded as an end-to-end analogue connection with defined loss, noise and bandwidth, etc. Therefore voice, fax and data sent over the PSTN are considered to be telephony.

Services can be anything from an e-mail service to an on-line share-dealing service. The on-line share dealing service may require such services as secure Web page hosting, e-mail, etc. Therefore a service provider may be a customer of other service providers.

Services can be categorised into two types — basic telecommunications services (such as telephony, e-mail, IN services) provided by network service providers, and higher value (content) services provided by high-level service providers. The high-value services need not be telecommunications services as such, but services that form part of other value chains that utilise the basic telecommunications services. A share-dealing service is not a telecommunications service, but may utilise some of the basic network services such as Web page hosting and e-mail.

The value-added by a service provider is difficult to specify as the role can cover any kind of service that exploits the network. However, higher value services generally involve value-adding by the sharing of a scarce resource. The scarce resource shared by the share-dealing service is the expertise (and the right) to buy and sell shares.

It is assumed that the service provider market may be much more dynamic than the infrastructural roles of network and access. Therefore scoping may concentrate on keeping one step ahead of the competition.

5. Topologies

A question arises as to how all these value-adding enterprises interact with each other and co-operate in order to meet the wide variety of needs of the customers. Figure 6 shows the elements that can be connected to form the topologies.

What will the supply chain look like? Clearly for the customer to be able to use the service they must be supplied, either directly or indirectly, with all the required elements, i.e. access, network and service. Hypothetically all the enterprises can supply each other and they can all supply the customer. A supply topology is characterised by who supplies whom within the supply chain, i.e. which of the possible supply conversations exists. There are many possible supply chain topologies. The question remains, which topology, or topologies, are most likely to prevail. An example topology is shown in Fig 7.

In this topology the customer has to procure all the elements required to use the service — thus all three enterprises supply direct to the customer. This means that the customer has to aggregate the three elements that make up the service. This topology puts an onus on the customer and leaves the customer to solve integration issues; therefore it is not a 'user-friendly' model. However, it is one that can be used to describe Internet services via dial-up

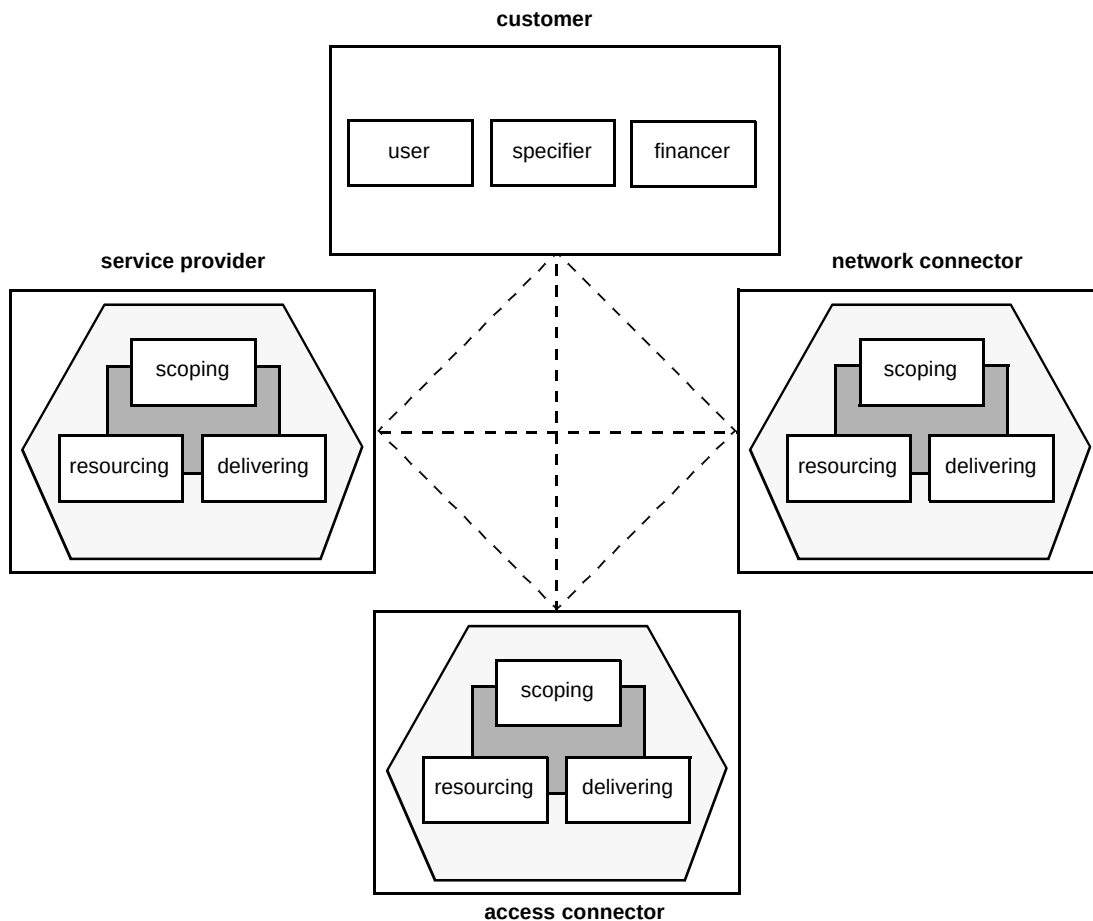


Fig 6 Topology elements.

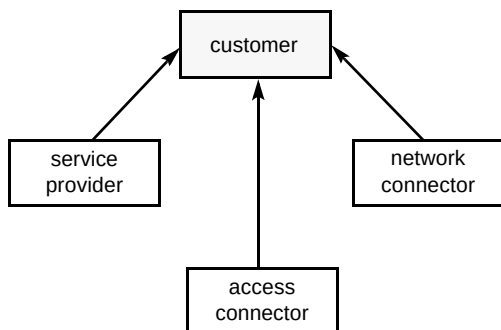


Fig 7 Customer aggregation topology.

Internet access². In this instance the access connection is provided by the PSTN operator (which is its own service-network-access supply chain), the network connector is the Internet, or more specifically the Internet service provider to whom the user subscribes, the service is supplied by the owner of the Internet-based service or site, e.g. Yahoo.

For most dial-up Internet users there are three contracts formed, and three different post-sales support contact points from which to choose. Which is the correct one to use requires the user to have some skill and knowledge of the

elements they have aggregated. The alternative is where the suppliers are willing and able to support the customer regarding problems they have with a service that they have not supplied, e.g. BT diagnosing faults that a customer has with their ISP.

In this topology there are no supply conversations between the access, network and service enterprises, although there may be conversations between the scoping parts of the enterprises so as to co-ordinate the enterprises' current and future offerings. The demand for the elements that each of them supplies is related to the capabilities and demand for the elements offered by the other two.

Other sets of interactions between the three supplying roles are likely. It is proposed that, of the many possible likely topologies, three will prevail, relating to three customer types:

- individual — residential customers and single-site businesses,
- SME — small-medium enterprises, i.e. businesses with a few sites and up to a few hundred employees,
- corporate — large international businesses.

² This is an interesting example as telephony access, network and service are being used to implement the access connector for the Internet.

The corporate model is not meant to imply that every corporate customer will procure all their telecommunications services in the same manner. These are illustrative, rather than definitive, models.

The corporate model is explained and analysed in the following section. Other topologies are then briefly described. The same conversation within the three topologies is analysed to illustrate the different characteristics of that conversation within the different topologies.

6. The corporate model of the telecommunications supply chain

Within the topologies, the broken line (in Fig 8) represents a supply conversation between the enterprises. The instruments for these conversations are not shown.

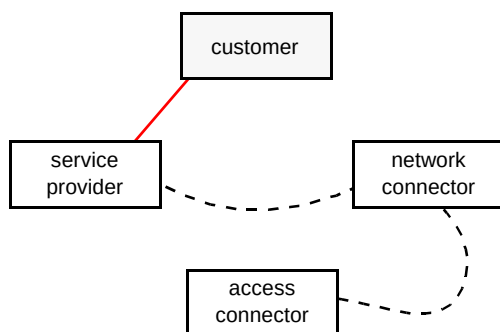


Fig 8 Corporate supply model.

The corporate supply model represents the spending power of the large global customer and its wish to have a 'one-stop-shop'. The corporate customer will need a range of services, network capacity and access connectivity throughout all its geographical locations. In order to maximise purchasing power and minimise purchasing effort, it will seek to procure all these services by conversing only with service providers. Indeed one of the services that the corporate customer will procure is a service to procure all the other services required. This is analogous to the currently fast-growing systems integration market, the 'solutions business'.

The service provider that faces the customer then procures the network connectivity it requires to deliver those services. In turn, as with the SME model, the network connector procures the required access connectivity to ensure that all the required locations are connected with the required bandwidth.

Within this model, the customer is not concerned with the revenue flows between the various enterprises, as the customer is only ever invoiced by the service provider. The service provider can charge a premium for this one-stop-

shop, as it reduces the purchasing and contract maintenance costs to the corporate customer.

As these models are generic, the analysis has to be general. Much more analysis is possible when specific instances are the subject. The specifics of a scenario will have great bearing in the findings of the analysis. Factors such as the type of service being offered, the value of the service being offered and the characteristics of the market for that service. However, the analysis in the following sections acts as an illustration.

6.1 Analysing the corporate supply model

Within the corporate supply model (Fig 8), the service supplier is the only enterprise which supplies direct to the end-customer. This means that the service supplier relies on the other enterprises to meet its customer's requirements.

There will be many different conversations that will take place between the enterprises, at various stages (epochs), e.g. conversations related to pre-sales, procurement, usage, post-sales. The procurement conversation will be the conversation that is analysed within this section.

6.2 Customer to service provider conversation

6.2.1 Mutuality

The customer in this case is buying a portfolio of services. This extensive range of requirements implies that the mutuality is high for the service provider; since they have taken on the responsibility to meet all the customer's requirements. There is very little onus on the customer to satisfy the supplier, other than by making their requirements clear and paying the bill; therefore the mutuality on the customer side of the conversation is low.

The customer may aim to apply some balance to the reliance it has placed with the service supplier, maybe through service level agreements and penalty clauses. The need for such devices will also be related to the significance of the conversation to the customer.

6.2.2 Significance

As the number of large global customers is relatively low, the significance to the service provider will be high. Each contract will represent a large proportion of the enterprise's annual turnover. The significance to the customer enterprise will also be high in that the services being procured may be vital to the running of their business, and a change of supplier would be costly and problematic. However, the monetary value of the service may only represent a small part of their total expenditure for the year,

and therefore the significance may be lower on the customer side of the conversation.

The customer will use the significance of its purchase as bargaining power in order to achieve the lowest possible price for the specified service. This significance may be reflected in the resulting conversation between the service provider and the network provider; the service provider will aim to use this as bargaining power.

6.2.3 Control

Control within this conversation lies very much with the customer. The conversation will probably have been initiated by the customer, via a tender or similar. The supplier will probably be competing against other suppliers to win the contract within a tender process defined by the customer. However, once the transaction is agreed, the control reverts to the service provider, who has control (within the terms of the contract) over how and when the work is executed within the terms of the contract.

6.2.4 Information

Prior to the procurement conversation, there will have been an exchange of some information between the enterprises, namely sales literature and advertising, so that the customer has enough information to determine a reasonable statement of their requirements. Once the procurement process is under way, most of the key information lies within the customer, e.g. budget, requirements. This richness of information presents a challenge to the service supplier by virtue of the need to ensure that these requirements are accurate and complete so as to ensure the satisfactory completion of the contract. This is also influenced by the mutuality of the conversation. The requirement on the service provider to satisfy its customers means that the provider will wish to know as much as possible about the requirements of the customer.

These values are summarised in Table 2. The values given to the metrics are restricted to high, medium or low.

Table 2 Metrics for the service provider to customer conversation.

	Customer	Service provider
Mutuality	low	high
Significance	high	high
Control	high	medium
Information	high	medium

Table 2 should be interpreted in the following way:

- mutuality — the customer has low mutuality, while the service provider has high mutuality for the conversation,
- significance — the conversation is highly significant to the customer and highly significant to the service provider,
- control — the customer has a lot of control of the conversation, while the service provider has some control, but not as much as the customer,
- information — the customer has a great deal of information to communicate to the service provider, while the service provider has a significant amount of information to pass to the customer.

6.3 Other conversations within the supply chain

This section looks at the other conversations within this model and uses the metrics of mutuality, significance, control and information to see how these conversations differ in nature from the one described above.

Table 3 summarises the conversational analysis of the conversations within the telecommunications corporate supply chain.

6.3.1 Mutuality

The high mutuality that the service supplier has for the customer is, in some sense, now represented in (inherited by) the service provider to network connector conversation. This is evident in the high mutuality that the network connector now has for the service provider. A failure of the network will cause a failure of the service. The network connector is not visible to the customer, who will blame all failures on the service provider. The same argument extends to the network connector to access connector conversation; any failure in the access connection will prevent the network connector from meeting its obligations to the service provider.

In the service supplier to network connector conversation and in the network connector to access connector conversation there is also some mutuality in the other direction. The network connector does not directly face the customer and therefore relies on service providers to sell service that consume network capacity. A failure to do this may lead to an under-utilisation of the network and consequential price increases. The same argument extends to the access connector relying on network connectors to sell their access capacity. It is in nobody's interests for enterprises in the supply chain to fail.

Table 3 Metrics for the conversations within the telecommunications supply chain model.

	Customer to service provider		Service provider to network connector		Network connector to access connector	
	Customer	Service provider	Service provider	Network connector	Network connector	Access connector
Mutuality	low	high	medium	high	medium	high
Significance	high	high	high	high	medium	medium
Control	high	medium	high	low	high	low
Information	high	medium	medium	low	medium	low

6.3.2 Significance

The significance to the network connector of the service provider to network connector conversation is high as the monetary value of the conversation will be high. As stated before there are only a limited number of these global customers and there will be a limited number of service providers in the solutions business.

The service provider sees significance inherited from its customer conversation, and relies on the network connector(s) to deliver to specification for a price that allows the service provider sufficient margin.

The significance within the network connector to access connector is lessened due to the presumed geographically limited coverage of the access connector. In order to meet a global customers' requirements, the network connector may need to form numerous contracts with many access connectors, some of which may be of very limited monetary value. This dilutes the significance of the conversation both to the access connector and the network connector.

6.3.3 Control

As with most customer-to-supplier conversations the control lies with the customer. This is especially true for conversations that have high supplier mutuality/significance.

Therefore the service provider will wish to control the conversation with the network connector, possibly requesting facilities such as on-line ordering, just-in-time provisioning and access to fault information as part of the procurement epoch. The network connector will aim to have some control within the conversation via terms and conditions and minimum length of contract, etc. This is to ensure that they can recoup costs incurred in the process of the contract set-up and to enable the generation of trust required to allow access to the network connector's systems.

The same arguments hold for the network connector to access connector conversation, with the network connector

looking for the access connector to be as responsive and accommodating as it is being to the service provider.

6.3.4 Information

As information passes from the service provider to network connector and then to access connector the same challenge of matching requirements to capabilities is present. The high mutuality implies that the supplier within each conversation will want to know as much as possible (that is relevant) about the service which their network (or access connection) is going to support to ensure that the customer's true requirements are met. This may include the prioritisation of connections so that fault repairs can be given appropriate precedence.

The potentially long lead times of supply for the infrastructural roles of network connector and access connector mean that they will want to know about the future requirements as well as current requirements of the service providers.

However, the nature of this information is far less rich than within the customer to service provider conversation, and this is reflected in the values given in Table 3.

7. Other supply chain topologies

It has been shown how different positions within the same supply chain create different types of conversation. This section now looks at how the same positions within different supply chains create different natures of conversation.

Two other supply chain topologies that are likely to exist are briefly described, together with how these topologies affect the network connector to access connector procurement conversations. Certain market assumptions have been made within these models to enable the analysis to be undertaken; when modelling specific supply chains more market data is required.

7.1 The SME model

It is assumed that the SME customers will have many sites with which they wish to interconnect, be they remote working employees, their customers or their suppliers.

The topology is shown in Fig 9. The customer procures network connectivity direct from the network connector and procures services direct either from the service provider or via the network connector.

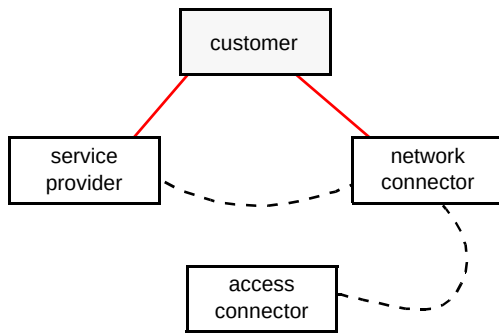


Fig 9 SME model.

The small and medium enterprise (SME) customer type is a specialist supplier within its own value chain. It is a business customer, interested in making improvements in its own operating efficiency or widening its market. As an SME it probably cannot afford (or does not perceive it as value for money) to have its entire network service requirements handled by a higher value service provider that delivers the whole package based on the SME's specifications.

The size and geographical spread of the SME makes them appropriate customers of the network connector. The SME may be so dispersed that a single access connector cannot meet all their needs. Indeed the company may be 'virtual' and totally reliant on the network to connect many points of presence. Thus the network connector can integrate the various access connector capabilities required.

7.2 Individual (residential) model

It is proposed that the model in Fig 10 is appropriate for the provision of services to single-site customers or individuals. Within this model the customer procures access direct from the access connector. The provision of network is either via a service provider or via the access connector, or both (e.g. some services may have specific network requirements when the service provider would pay for the transmission of the content and pass the cost on to the customer). In certain circumstances the access connector may buy in network capacity and telecommunications services for re-sale to the end customer; however, for higher value services the customer may go direct to the service provider.

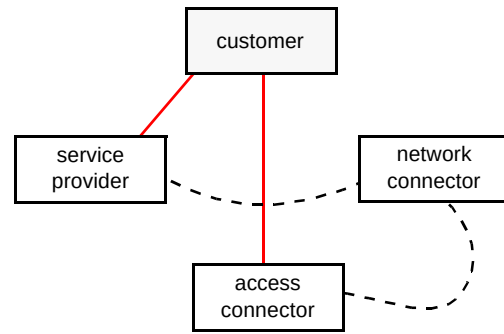


Fig 10 Individual customer topology.

These users may access services from places other than their home or business premises³, but are going to be responsible for attaining access to the network for their own property (or mobile terminal). The access that the individual procures will dictate the range of services they can use. The access channel(s) procured by the customer will be shared by various services, in a similar way to a PC running many applications.

As the access connector will form part of many value chains, it is very important to the customer to seek the access service that suits them best. An overpriced or underperforming access connection will impact on the value they receive from all the services they consume. Therefore the individual will want to deal directly with the access connector.

As access technologies provide greater capability to the customer, there may be a range of access offerings and access connectors from which to choose. It seems likely, due to regulation and technology, that competing access connectors and network connectors will be able to choose and change with whom they interconnect.

7.3 Network connector to access connector conversations

Table 4 summarises the conversational analysis of the conversations within other supply chain topologies.

7.3.1 Mutuality

Within the SME model the access connector is a supplier to the network connector, just as it was within the corporate model. Therefore the network connector relies on the access connector to meet its customers' requirements. Again, as the access connector does not sell direct to the

³ For example, current technology allows people to use BT Chargecards, and therefore in theory you could make all your calls via a chargecard, negating the need to pay for access to your residence. However, this is less than convenient. Therefore individual customers require access for their homes even if they also use access infrastructure that other people have paid for.

Table 4 Metrics for the conversations within the model for other supply chain topologies.

	Corporate model		SME model		Individual model	
	Network connector	Access connector	Network connector	Access connector	Network connector	Access connector
Mutuality	medium	high	medium	high	high	low
Significance	medium	medium	medium	high	medium	high
Control	high	low	high	medium	low	high
Information	medium	low	medium	low	low	low

customer, they rely on the network connector(s) to sell their access connections for them.

Conversely, within the individual model, the access connector faces the customer and may even 'badge' the network connectivity. This reverses the mutuality within the conversation as the access connector is now heavily reliant on the network connector. The customer will perceive poor network performance as poor service from the access connector.

Network connectors rely on access connectors to connect customers to their network, but there is less reliance than in previous models, and hence low mutuality. This is because the network connectors gain revenue from the use of services, and therefore some mutuality is with the service provider, the network connectors relying on them to deploy services that generate network demand. Additionally network connectors can also gain revenue from other network connectors, via interconnection agreements. Therefore the end customer is not their sole direct source of revenue.

7.3.2 Significance

Within the corporate model it is assumed that the geographical scope of a corporate customer is so great that a significant number of access connectors are required to meet their needs. The SME is also a dispersed organisation, but the scope of this dispersal is such that their requirements can be met by one or two access connectors. Therefore SMEs may be more valuable than global companies as customers of access connectors. This is reflected within the SME model in the high significance for the access connector of its conversation with the network connector.

Within the individual model, a customer may only procure access from one access connector (or one mobile and one fixed access connector). Therefore the access connector becomes the only sales channel for the network connectors who wish to sell capacity to that customer. However, each conversation only relates to an individual, and therefore corresponds to a relatively small consumption of network capacity and correspondingly low monetary value.

Assuming that the access connector procures network capacity for re-sale to their customer, the conversation that procures that network capacity is high. The price, quality and capability of the access and network bundle that the access connector offers its customers will depend heavily on those characteristics of the network procured.

7.3.3 Control

The network connector in the SME model is a customer of the access connector. In the individual model the roles are reversed, and the network connector supplies the access connector. This is reflected in the values for control in Table 4, the customer having more control in the conversation than the supplier. The access connector has more control as a supplier than the network connector on the assumption that the access connector may own the only access connection to a customer's premises.

7.3.4 Information

Within the SME and individual models there needs to be very little information passed between the two enterprises in this conversation. The two enterprises may choose to share more information than the minimum, e.g. advance notice of access requirements may enable the access connector to reduce lead times.

The 'low' grading for the individual model reflects the assumption that less information is needed to satisfy an individual customer than an SME.

8. Discussion

The four dimensions of conversation theory are not totally orthogonal. Table 5 shows how high customer significance and supplier mutuality may influence the other metrics within that conversation.

Table 5 Metric for a conversation with high customer significance and high supplier mutuality.

	Mutuality	Significance	Information	Control
Customer	low	high	high	high
Supplier	high	??	high	low

Where a cell indicates high mutuality, the supplier needs to undertake responsibilities on the customer's behalf. This requires the supplier to understand what those responsibilities are and this implies a high degree of information.

Similarly, high customer significance implies the need to ensure that the conversation succeeds, again implying high information and the wish for customer control of the conversation. For example, people will spend a long time surveying the market prior to the purchase of an expensive item such as a car or house. The significance of the conversation to the customer and their reliance on the supplier leads to the customer wanting to control the conversation as much as possible. High customer significance and mutuality tends to imply significance to the supplier, though if a supplier has many customers, each individual conversation will not necessarily assume great significance.

Such generalisations may only hold true for a certain type of supply conversation. The nature of a supply conversation between a wholesaler and a retailer is different from that between a supplier and an end customer. Work is under way to identify and characterise the different types of supply conversation in terms of the value of, and relationship between, the four conversational dimensions.

The nature of the supply chain can have a dramatic effect on the nature of the interaction between any two roles. The mutuality, significance, information, and control can describe the conversation. Where an enterprise boundary has high significance and mutuality on both sides the enterprises will look to co-operate — they rely on each other and will wish to reduce the risk of failure. Sharing information and providing more control to the other party may result. One way of doing this is to share systems or processes, such as ordering and procuring systems, problem management systems. Such systems would require open interfaces and data models that accommodate this openness.

However, business environments change, and change rapidly, and co-operation can turn to competition. The significance may reduce for one or both enterprises, or business policy may be altered for other reasons. Interaction may need to be terminated in the future and designing the interfaces is a balancing act. The value-adding model and the high-level supply chain models are necessary but not sufficient. Detail is required to understand what contributes to the significance and mutuality and what constitutes the information and what is controlled. A detailed model for service supply, and the associated responsibilities for supporting a service company, is needed. From this model the design of systems and processes can start. This is an area of continuing research.

9. Conclusions

A language for an enterprise modelling method has been described which incorporates the ability to assess the nature of the interaction between two roles at a business level. A new model for a value-adding enterprise has been defined. A value-adder has autonomous control over scoping, resourcing and delivery responsibilities. Three types of value-adder role have been described within the telecommunications industry, the service provider, the network connector and the access connector. A conversation theory using dimensions of mutuality, significance, information and control has been applied to compare the role interactions in three likely telecommunications supply chains (the corporate, the SME and the individual).

The analysis has shown how critical the supply chain topology can be to the nature of inter-role conversations. The analysis uses roles and not players as an essential feature. The analysis method presented is very useful when assessing which role to adopt in a market and evaluating the scenarios and strategy options. This coupling of high-level enterprise models (that can be expanded for specific situations) with conversation theory enables consistent viewpoints to be generated and scenarios compared.

Since the nature of the role interaction will dictate the requirements of the operational support systems and processes, the models described are important pieces in strategy analysis. Further work on service-supplying models will enable a full description of the operational support options for each role. Ongoing research will progress this area.

References

- 1 Russell C M, Holladay M R and Barnsley P B: 'The Future World of Telecommunications', *British Telecommunications Eng J*, **17**, No 3 (October 1998).
- 2 Whitham L E and Huff B L: 'A Living Enterprise Model', *Proc 6th Industrial Engineering Research Conf* (May 1997).
- 3 Fox M and Gruninger M: 'Enterprise Modeling', *AI Magazine*, **19**, No 3, pp 109—121 (Fall 1998).
- 4 Liles D H and Presley A R: 'Enterprise Modeling within an Enterprise Engineering Framework', *Proc 1996 Winter Simulation Conf*, pp 993—999 (December 1996).



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